

LECTURE #6

Econometrics I

TESTING MULTIPLE LINEAR RESTRICTIONS & OLS ASYMPTOTICS

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In the previous lecture #5

- ▶ We discussed the **Gauss-Markov theorem**: OLS is **BLUE**.
- ▶ We added **MLR.6 Normality**: OLS is **BUE** and

$$\begin{aligned}y|X &\sim N(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k, \sigma^2), \\ \hat{\beta}_j &\sim N(\beta_j, \text{Var}(\hat{\beta}_j)).\end{aligned}$$

- ▶ We introduced the **t test**: under MLR.1–6 and $H_0 : \beta_j = a_j$,

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j - a_j}{\text{se}(\hat{\beta}_j)} \sim t_{n-k-1} = t_{df}.$$

- ▶ $1 - \alpha$ **confidence interval** for β_j : $\hat{\beta}_j \mp t_{n-k-1, 1-\alpha/2} \text{se}(\hat{\beta}_j)$.
- ▶ Readings for lecture #6:
 - ▶ Chapter 4: 4.5, 4.6, Chapter 5.

Next week: Midterm test

- ▶ No seminars.
- ▶ Midterm test instead of lecture:
 - ▶ 65-minute, written, closed-book; 0–30 points, results via SIS,
 - ▶ **March 31, 11:00**, lecture hall MM 109,
 - ▶ no calculators, formula sheets, own blank papers, statistical tables, or vocabularies can be used; a pen fully suffices,
 - ▶ please bring your ID card (ISIC, national ID, passport),
 - ▶ see *Midterm_JEB109_2026_Instruction+Topics+Tables.pdf* in SIS.
- ▶ Midterm structure:
 1. multiple-choice questions,
 2. empirical exercise: interpretation of regression results, functional forms (logs, quadratic), data scaling, inference (t test, CI, F test), R^2 , expected bias analysis,
 3. theoretical derivations topics: OLS (MM or LS), OLS algebraic properties, 'sums of squares', unbiasedness (SLR, MLR), variance (SLR, MLR), linear estimator, F test (SSR vs. R^2).

Outline

Testing multiple linear restrictions

OLS asymptotics: Consistency

Asymptotic normality, efficiency, and large sample inference

Asymptotic normality and efficiency of OLS

Lagrange multiplier test

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Testing exclusion restrictions

- ▶ Up until now, we have been testing only single hypotheses.
- ▶ However, we might be interested in testing a set of multiple hypotheses/restrictions.
- ▶ Null hypothesis can then have the following form:

$$H_0 : \beta_1 = 0, \beta_2 = 0$$

against the alternative

$$H_1 : H_0 \text{ does not hold.}$$

- ▶ H_1 thus holds if at least one of β_1 and β_2 is non-zero (i.e., it complements the null).
- ▶ t statistic is not appropriate here, as we need to test the restrictions **jointly** (vs. multiple **separate** restrictions).

Unrestricted and restricted models

- ▶ Let us have an original model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u.$$

- ▶ As there are no restrictions, the model can be referred to as the **unrestricted model**.
- ▶ Using the restrictions from H_0 on the previous slide, we obtain the **restricted model**

$$y = \beta_0 + \beta_3 x_3 + u.$$

- ▶ For each, we can obtain SSR or the coefficient of determination R^2 to judge their performance, i.e., how well the models explain the variation in the dependent variable y .
- ▶ Recall that SSR of the unrestricted model SSR_U will **always be lower** than SSR of the restricted model SSR_R . Why?

F test

- ▶ Has SSR_R increased enough to allow us to reject H_0 ?
- ▶ In other words, has the model changed so much that the two models are statistically (based on SSR or R^2) distinguishable?
- ▶ To answer this question, we use the F **statistic** defined as

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} = \frac{(R_U^2 - R_R^2)/q}{(1 - R_U^2)/(n - k - 1)},$$

where $n - k - 1$ is the degrees of freedom of the original unrestricted model, and q is the number of restrictions.

- ▶ $F \geq 0$ (**always** non-negative!)
- ▶ Under MLR.1 through MLR.6, if the null hypothesis holds,

$$F \sim F_{q, n-k-1},$$

where q and $n - k - 1$ are the two degrees of freedom.

- ▶ Why is F distributed this way?

F test: SSR vs. R^2 form of the statistic

- ▶ Using the definition of $R^2 = 1 - \frac{SSR}{SST}$, we can rearrange to get $SSR = (1 - R^2)SST$.
- ▶ This can be used both for the unrestricted and restricted model.
- ▶ Keeping in mind that SST is the same for both models, as the dependent variable remains the same, we have

$$\begin{aligned} F &= \frac{SSR_R - SSR_U}{SSR_U} \frac{n - k - 1}{q} = \\ &= \frac{(1 - R_R^2 - 1 + R_U^2)SST}{(1 - R_U^2)SST} \frac{n - k - 1}{q} = \\ &= \frac{R_U^2 - R_R^2}{1 - R_U^2} \frac{n - k - 1}{q} = \frac{(R_U^2 - R_R^2)/q}{(1 - R_U^2)/(n - k - 1)}. \end{aligned}$$

F test

- ▶ Recall how to test a hypothesis utilizing the F distribution.
- ▶ How do we obtain the p -value for the test?
- ▶ If the null hypothesis is rejected, we say that the variables of interest are **jointly statistically significant**, ceteris paribus.
- ▶ If the null hypothesis is not rejected, we say that the variables of interest are **jointly insignificant**.
- ▶ In addition, F test can be used as one of the variable selection/elimination criteria to drop insignificant explanatory variables from the model.

F statistic for testing overall significance of a regression

- ▶ Econometric software usually reports the **overall F statistic** to complement the coefficient of determination R^2 as a measure of the estimated model's overall explanatory power.
- ▶ Statistical significance of R^2 is usually not provided.
- ▶ Such F statistic is used to test a joint hypothesis that **all model parameters but intercept** are equal to zero, i.e.,

$$H_0 : \beta_1 = 0, \dots, \beta_k = 0.$$

- ▶ I.e., we test so-called **overall significance of a regression**.
- ▶ Note that in this specific case, R^2 of the restricted model is 0 (as there are no explanatory variables) and $q = k$.
- ▶ Passing the F test for overall significance is considered the very minimum for the model quality judgment.

Testing general linear restrictions

- ▶ Frequently, we are interested in a more complicated joint hypothesis containing general linear restrictions, i.e., not only the 'zero excluding restrictions'.
- ▶ Procedure remains the same but often requires some creativity.
- ▶ Let us have the **unrestricted** model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + u.$$

- ▶ A set of linear restrictions $H_0 : \beta_1 = 1, \beta_2 = 0, \beta_3 = 0, \beta_4 = 0$ gives the **restricted** model

$$\begin{aligned} y &= \beta_0 + x_1 + u, \\ y - x_1 &= \beta_0 + u. \end{aligned}$$

- ▶ We can now only use the *SSR* form of the F statistic! Why?

CAPM model

- ▶ Interesting example: capital asset pricing model (CAPM).
- ▶ Model is specified as

$$r_{i,t} - r_{rf,t} = \alpha_i + \beta_i(r_{M,t} - r_{rf,t}) + u_{i,t}$$

with an interesting null hypothesis of $H_0 : \alpha_i = 0, \beta_i = 1$ of an efficient market hypothesis-following asset.

- ▶ If we substitute for H_0 , we obtain

$$r_{i,t} - r_{rf,t} = (r_{M,t} - r_{rf,t}) + u_{i,t},$$

so there is nothing to estimate!

- ▶ We thus take that as if it was estimated:

$$SSR_R = \sum_{t=1}^T \hat{u}_{i,t}^2 = \sum_{t=1}^T (r_{i,t} - r_{M,t})^2.$$

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Consistency

- ▶ Unbiasedness cannot always be achieved.
- ▶ Consistency is thus considered a minimal requirement for an estimator.
- ▶ In simple terms, an estimator is considered consistent if it gets **closer** to and **more tightly** distributed around the true population parameter with increasing sample size n .
- ▶ For $n \rightarrow +\infty$, a consistent estimator collapses to a single point β_j , i.e., to the true value of the population parameter.
- ▶ Under assumptions MLR.1 through MLR.4, the **OLS estimator $\hat{\beta}_j$ is consistent** for all $j = 0, 1, \dots, k$.
- ▶ This can be shown utilizing LLN (in seminar #6 for $\hat{\beta}_0$).
- ▶ Consistency vs. unbiasedness.

Consistency

- ▶ MLR.5 and MLR.6 are thus not needed for consistency, in the same way as for unbiasedness.
 - ▶ In addition, regarding MLR.4, we only need zero correlation (not mean independence) between explanatory variables and the error term for consistency.
- ⇒ **MLR.4' Zero mean and zero correlation:** $\mathbb{E}(u) = 0$ and $\text{Cov}(x_j, u) = 0$ for $j = 1, 2, \dots, k$.
- ▶ Inconsistency of $\hat{\beta}_1$ for the simple regression case is given by

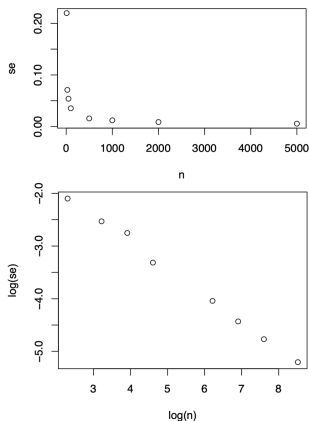
$$\text{plim}(\hat{\beta}_1) - \beta_1 = \frac{\text{Cov}(x_1, u)}{\text{Var}(x_1)},$$

which is parallel to the derivation of the bias, only the inconsistency is expressed in population terms.

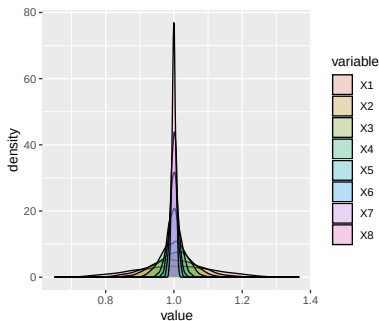
Consistency: R example

- ▶ $n \in \{10, 25, 50, 100, 500, 1000, 2000, 5000, (10k, 30k, 100k)\}$
- ▶ $y = 1 + 1x + u$

(a) convergence of se



(b) approximated $\hat{\beta}_1$ densities



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Asymptotic normality of OLS

- ▶ Normality of u is a very strong assumption, but it is needed for normality of the OLS estimator and, in turn, also for t -distributed t statistics and F -distributed F statistics.
- ▶ Fortunately, CLT comes to the rescue when MLR.6 is violated.
- ▶ Under the Gauss-Markov assumptions MLR.1 through MLR.5:
 - ▶ $\hat{\beta}_j$ is **asymptotically normally distributed** for each j :

$$\sqrt{n}(\hat{\beta}_j - \beta_j) \overset{a}{\sim} N(0, \text{asymptotic Var}_j),$$

- ▶ $\hat{\sigma}^2$ is a **consistent** estimator of $\sigma^2 = \text{Var}(u)$,
- ▶ and finally:

$$\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \overset{a}{\sim} N(0, 1) \quad \text{and} \quad \frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \overset{a}{\sim} N(0, 1).$$

Asymptotic normality and efficiency of OLS

- ▶ That means, if the sample is large enough and the Gauss-Markov assumptions MLR.1 through MLR.5 are met, **we do not need MLR.6** for statistical inference based on t and F statistics and the confidence intervals.
- ▶ These are referred to as the **asymptotic t statistics**, **asymptotic F statistics**, etc.
- ▶ We still need MLR.5 Homoskedasticity!
- ▶ In addition, under the Gauss-Markov assumptions MLR.1 through MLR.5, the OLS estimators are **asymptotically efficient**, i.e., the estimators with the lowest **asymptotic variances**.
- ▶ $\widehat{\text{Var}}(\hat{\beta}_j)$ shrinks to zero at a rate of $1/n \Rightarrow se(\hat{\beta}_j)$ at $1/\sqrt{n}$.
- ▶ How many observations do we need for the asymptotics?

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Lagrange multiplier test (for q exclusion restrictions)

- ▶ ***LM* statistic/test** is an alternative approach for testing multiple exclusion restrictions without assuming MLR.6.
- ▶ *df* play no role bcs. of the asymptotic nature of the statistic.
- ▶ Basic intuition: if the 'restricted' independent variables are truly irrelevant, then these should not be correlated with residuals of the restricted model.
- ▶ This recalls MLR.4 as the 'specification assumption'.

Lagrange multiplier test

LM test is constructed in the following steps:

1. H_0 and H_1 are the same as for the respective F test, e.g.:

$$H_0 : \beta_1 = 0, \beta_2 = 0 \quad \text{vs.} \quad H_1 : H_0 \text{ does not hold.}$$

2. Estimate the **restricted model** and save the residuals $\tilde{u}$.
3. Run an **auxiliary regression**: regress $\tilde{u}$ on **all independent variables** and obtain R^2 of this regression, i.e., $R_{\tilde{u}}^2$
(intuition: if H_0 is true, $R_{\tilde{u}}^2$ is 'close' to zero),
4. Compute $LM = nR_{\tilde{u}}^2$.
5. Under the null hypothesis, $LM \stackrel{a}{\sim} \chi_q^2$.
6. If $LM > c$, we reject H_0 at the given significance level α .

Seminars and the next lecture

- ▶ Seminars:
 - ▶ testing multiple linear restrictions: F test,
 - ▶ consistency derivation for $\hat{\beta}_0$,
 - ▶ normality of residuals and asymptotic variance in practice,
 - ▶ Lagrange multiplier test.
- ▶ Next lecture #7 (in two weeks):
 - ▶ midterm summary,
 - ▶ effects of data scaling on OLS statistics,
 - ▶ more on functional forms: quadratic, logarithmic, interactions,
 - ▶ more on goodness-of-fit: adjusted R-squared.
- ▶ Readings for lecture #7:
 - ▶ Chapter 6: 6.1–6.3.

Historical spotlight: Ronald Fisher (1890–1962)

Ronald A. Fisher is often called the founder of modern 20th-century statistics. He worked at University College London and Cambridge. His signature tool is the F test: compare how much groups differ with how much individuals vary inside each group. If the between-group variation is big relative to the within-group variation, the F statistics is large and the effect looks real. This is the logic of ANalysis Of VAriance.

Fisher liked simple, sharp demonstrations. In the 'lady tasting tea' story, a colleague said she could tell whether milk or tea was poured first. Fisher prepared eight cups, four each way, and mixed them at random. He asked her to sort them blindly. Probability gave the scoring rule: if she could do this better than chance, the claim had support. The lesson was not about tea. It was about careful planning of a randomized test, and a clear rule for deciding what counts as evidence.

He was also a principal founder of population genetics. He set out Fisher's principle on 1:1 sex ratios, the 'sexy-son hypothesis' in sexual selection, or pioneered gene mapping. Takeaway for econometrics: plan before you estimate, utilize clear decision rules, and use the F test to compare groups and models. F tests help decide whether an observed difference is real or just noise, especially when several treatments or factors are in play.



Source: [Wikipedia](#)